Assignment:16.2

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Course:Ai coding

Batch:33

Task Descrption-1: (Design Database Schema for Student Management System)

Instructions:

• Using AI tools, generate CREATE TABLE statements for a Student Management Database.

• Tables: Students, Courses, Enrollments.

• Include primary keys, foreign keys, and suitable data types. Ensure the schema follows normalization rules (1NF, 2NF, 3NF).

Sample Code:

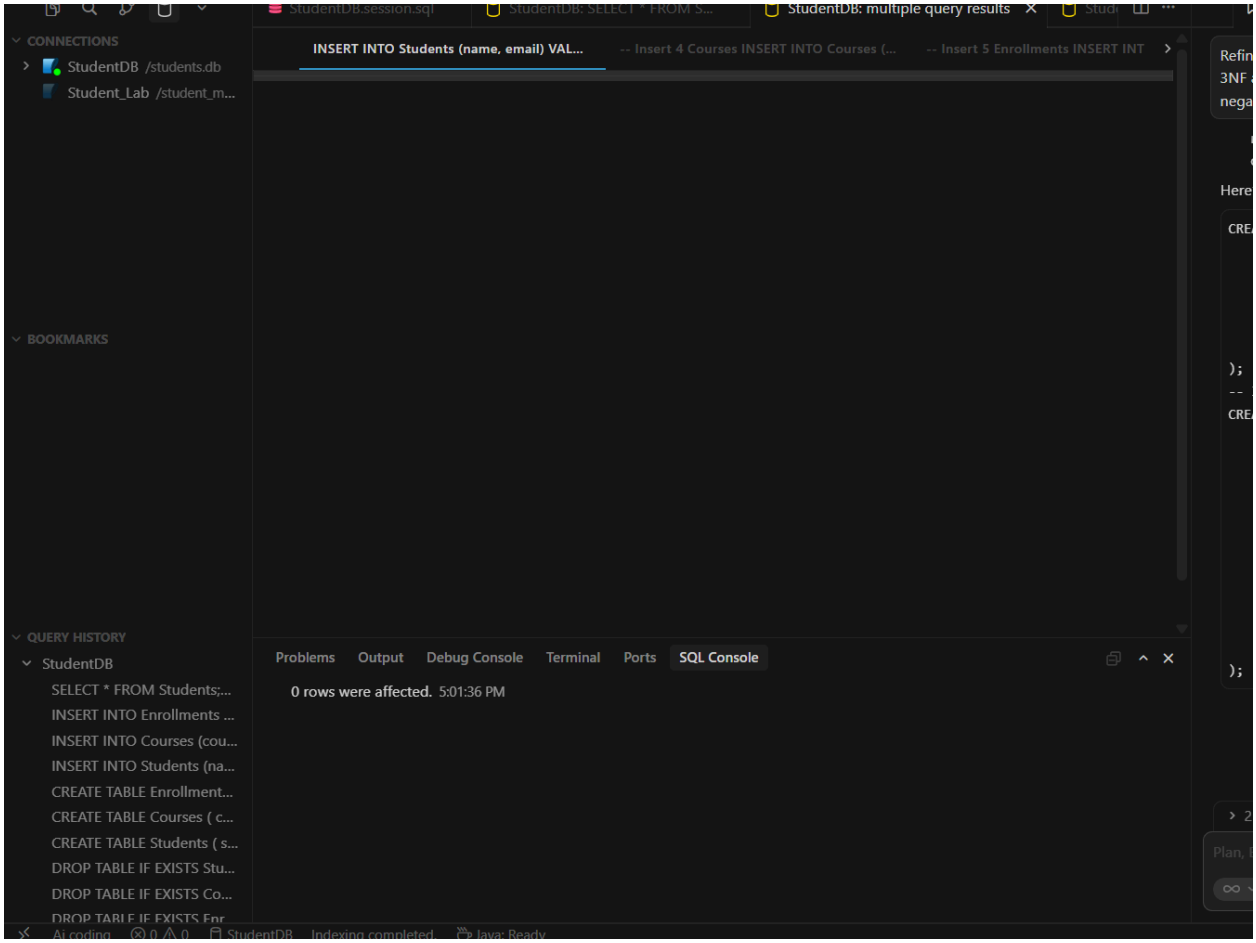
```
CREATE TABLE Students (
student_id INT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100)
);
CREATE TABLE Courses (
course_id INT PRIMARY KEY,
course_name VARCHAR(100),
credits INT
);
CREATE TABLE Enrollments (
enrollment_id INT PRIMARY KEY,
student_id INT,
```

course_id INT,
FOREIGN KEY (student_id) REFERENCES Students(student_id),
FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);

Expected Output:

Tables Students, Courses, and Enrollments are created successfully.

Foreign key constraints are properly enforced.



Task Description- 2: (Insert Sample Records)

Instructions:

• Use AI tools to generate INSERT statements to populate the tables.

• Insert at least 3 students, 4 courses, and 5 enrollment records. Verify data using SELECT queries.

Sample Code:

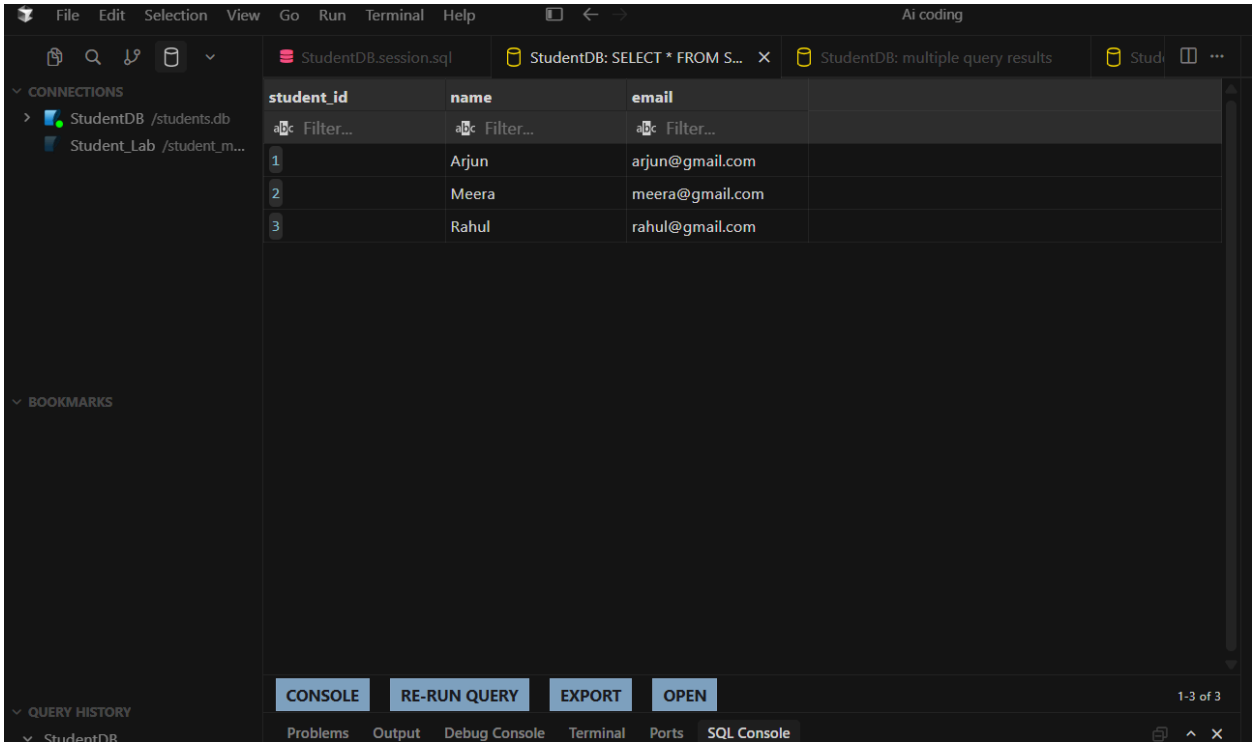
```
INSERT INTO Students VALUES
(1, 'Arjun', 'arjun@gmail.com'),
(2, 'Meera', 'meera@gmail.com'),
(3, 'Rahul', 'rahul@gmail.com');
INSERT INTO Courses VALUES
(101, 'Database Systems', 4),
(102, 'Data Structures', 3),
(103, 'Operating Systems', 4),
(104, 'AI Fundamentals', 3);
INSERT INTO Enrollments VALUES
(1, 1, 101),
(2, 1, 102),
(3, 2, 103),
(4, 3, 101),
(5, 3, 104);
```

Expected Output:

• Sample records are inserted successfully.

• FROM Students; displays all student records.

• SELECT * FROM Enrollments; displays all enrollments.



Task Description- 3: (Perform CRUD Operations)

Instructions:

Use AI to generate SQL queries for basic CRUD operations:

- Retrieve all courses
- Update a student's email
- Delete an enrollment record
- Validate the changes using SELECT statements.

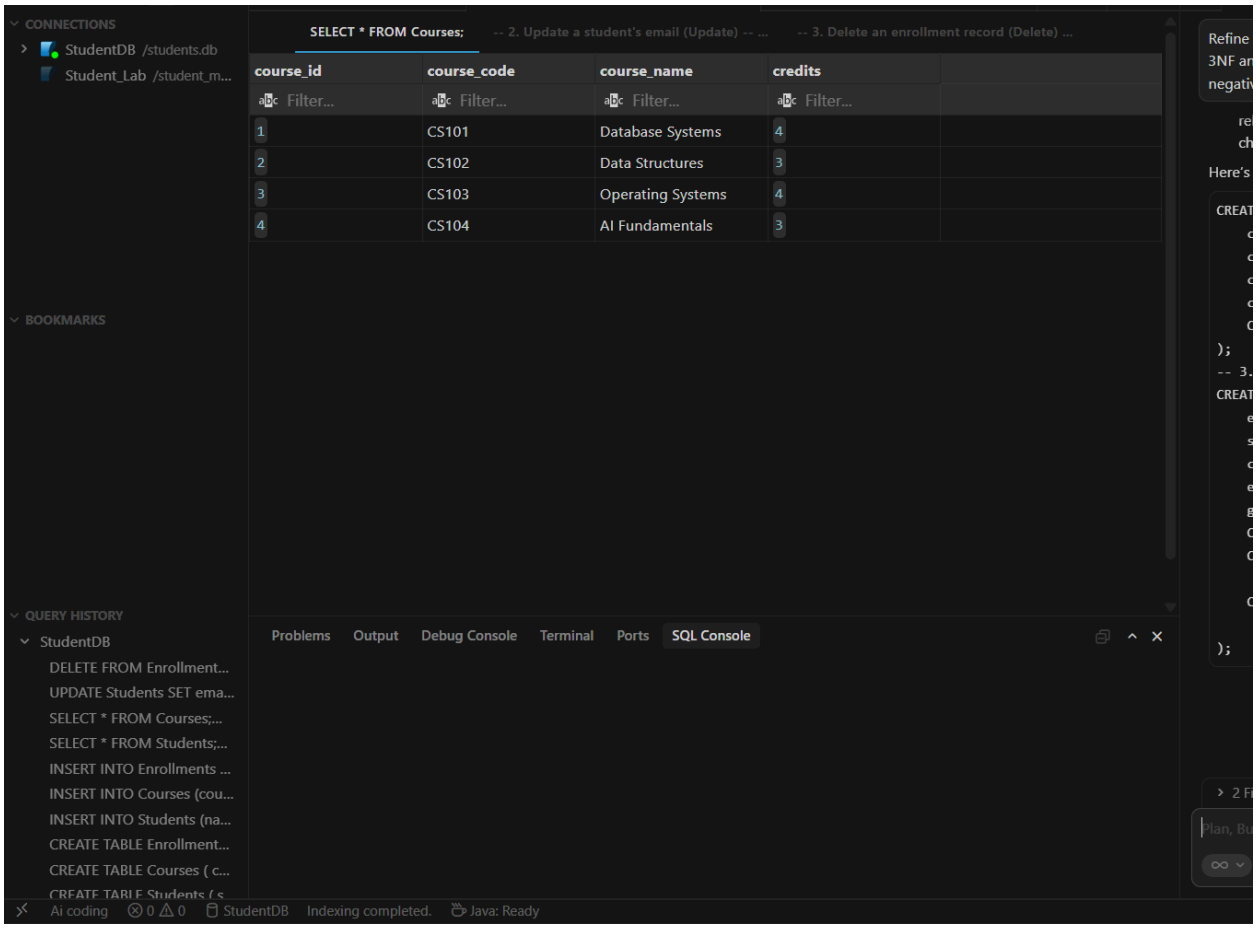
Sample Code:

```
SELECT * FROM Courses;
UPDATE Students
SET email = 'arjun_new@gmail.com'
WHERE student_id = 1;
DELETE FROM Enrollments
WHERE enrollment_id = 5;
```

Expected Output:

- Updated email is reflected in Students table.
- Deleted enrollment record is removed successfully.

• Remaining data remains unchanged.



Task Description- 4: (Execute Aggregate Queries)

Instructions:

Use AI tools to write aggregate queries:

• Count number of students enrolled per course

• Display courses with more than one student enrolled

Analyze the correctness of results.

Sample Code:

SELECT c.course_name, COUNT(e.student_id) AS student_count

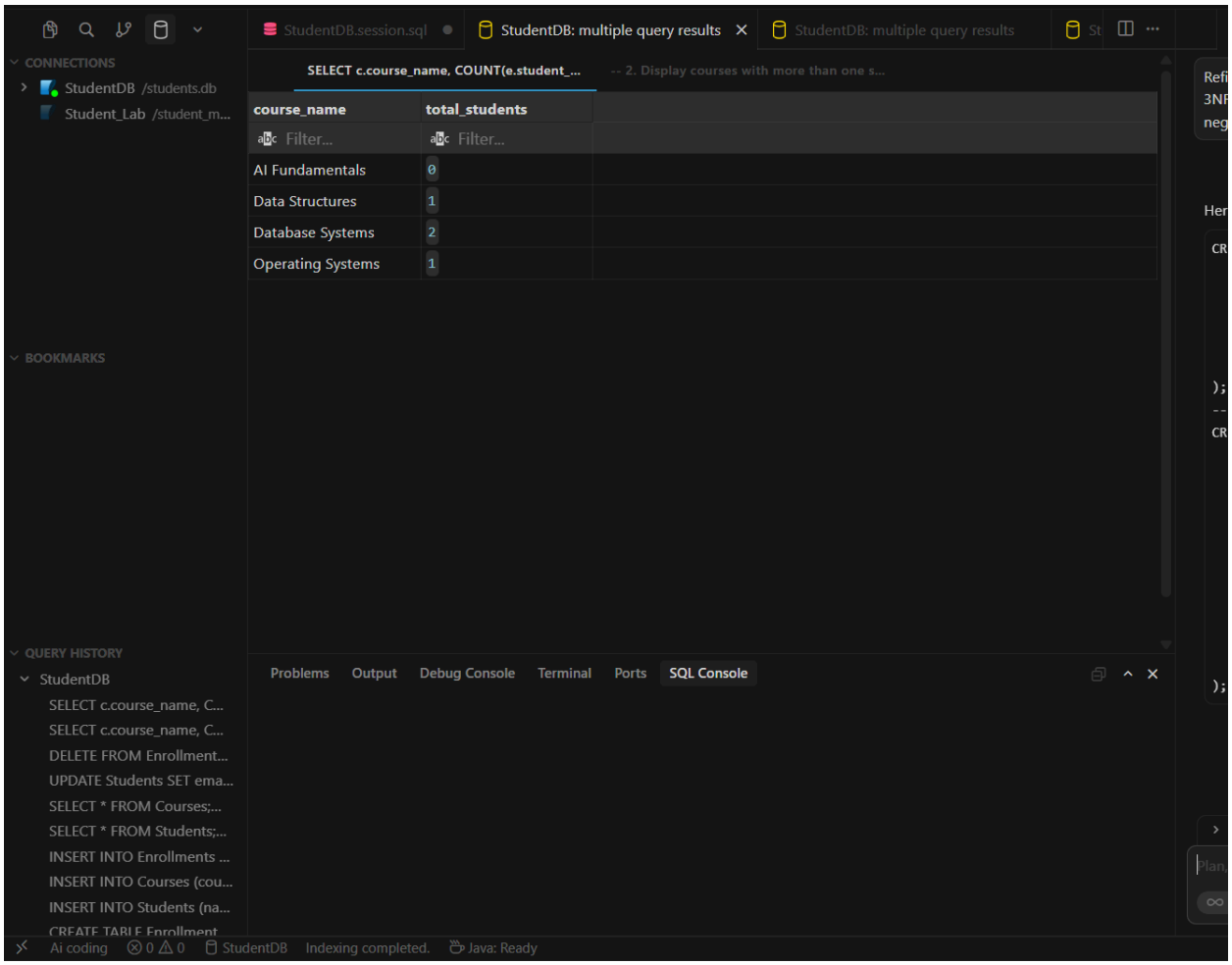
FROM Courses c

JOIN Enrollments e ON c.course_id = e.course_id

GROUP BY c.course_name;

SELECT c.course_name

FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_name
HAVING COUNT(e.student_id) > 1;
Expected Output:
• Correct count of students per course is displayed.
• Courses with multiple enrollments are listed accurately.



Task Description- 5: (Query Optimization)
Instructions:
• Use AI tools to analyze query performance.
• Rewrite inefficient sub queries using JOINs and test both queries
for correctness.

Sample Code:

```
-- Original Query
SELECT * FROM Students
WHERE student_id IN (
SELECT student_id FROM Enrollments WHERE course_id = 101
);
-- Optimized Query
SELECT s.*
FROM Students s
JOIN Enrollments e ON s.student_id = e.student_id
WHERE e.course_id = 101;
```

Expected Output:

- Both queries return the same set of students.
- Optimized query performs better on large datasets.

This lab allows students to use AI tools like Copilot, Cursor, or Gemini to:

- Generate schema definitions
- Auto-complete insert statements
- Write CRUD and aggregate SQL queries
- Optimize queries and ensure proper normalization

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Ai coding

StudentDB: multiple query results | StudentDB: multiple query results | StudentDB: REATE TABLE IF N...

CONNECTIONS

- StudentDB /students.db
- Student_Lab /student_m...

BOOKMARKS

QUERY HISTORY

student_id	name	email
Filter...	Filter...	Filter...
1	Arjun	arjun_new@gmail.com
3	Rahul	rahul@gmail.com

CONSOLE | RE-RUN QUERY | EXPORT | OPEN

1-2 of 2

CONNECTIONS

- StudentDB /students.db
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QUERY HISTORY

- StudentDB

Problems | Output | Debug Console | Terminal | Ports | SQL Console